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CSE 6015, Statistical Machine Translation

Lecture sheet-7

Computing Model 1 parameters:

How to compute translation probabilities for model 1 from a parallel corpus?

Step1: Determine candidates. For each English word e collect all foreign words f that co-occur at least once with e.

Step2: Initialize $P(f|e)$ uniformly

$$P(f|e) = \frac{1}{\text{No of co-occuring foreign words}}$$

Step3: Iteratively refine translation probabilities

```
for n iterations
    set tc to zero
    for each sentence pair (e, f) of lengths (l, m)
        for j = 1 to m
            total = 0
            for i = 1 to l
                total = total + P(f_j | e_i)
            end for
            for i = 1 to l
                tc(f_j | e_i) = tc(f_j | e_i) + P(f_j | e_i) / total
            end for
        end for
    end for
    for each word e
        total = 0
        for each word f
            total = total + tc(f|e)
        end for
        for each word f
            P(f|e) = tc(f|e) / total
        end for
    end for
end for
```

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Example: (parallel corpus)

the dog :: le chien

the cat :: le chat

step 1+2 (collect candidates and initialize uniformly)

the → le, chien, chat

dog → le, chien, chat

cat → le, chat, chien

NULL → le, chien, chat

$$P(le|the) = P(chien|the) = P(chat|the) = \frac{1}{3}$$

$$P(le|dog) = P(chien|dog) = P(chat|dog) = \frac{1}{3}$$

$$P(le|cat) = P(chat|cat) = P(chien|cat) = \frac{1}{3}$$

$$P(le|NULL) = P(chien|NULL) = P(chat|NULL) = \frac{1}{3}$$

step 3: Iterate

e_1 NULL e_2 the e_3 dog :: f_1 le f_2 chien

e, $l = 3$; f, $m = 2$

$j = 1$ total $= 0$ $i = 1$ total $= 0 + P(f_1|e_1)$

$i = 2$ total $= P(f_1|e_1) + P(f_1|e_2)$

$i = 3$ total $= P(f_1|e_1) + P(f_1|e_2) + P(f_1|e_3)$

$= P(le|NULL) + P(le|the) + P(le|dog)$

$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

$i = 1$ $tc(f_1|e_1) = 0 + \frac{P(f_1|e_1)}{total} =$

$\Rightarrow tc(le|NULL) = P(le|NULL)/1 = \frac{1/3}{1} = 0.33$

$i = 2$ $tc(le|the) = P(le|the)/1 = \frac{1/3}{1} = 0.333$

$i = 3$ $tc(le|dog) = P(le|dog)/1 = \frac{1/3}{1} = 0.333$

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$j = 2$ total = 0 $i = 1$ total = $0 + P(f_2|e_1) = P(f_1|e_1)$

$i = 2$ total = $P(f_2|e_1) + P(f_2|e_2)$

$i = 3$ total = $P(f_2|e_1) + P(f_2|e_2) + P(f_2|e_3)$

$= P(\text{chien}|\text{NULL}) + P(\text{chien}|\text{the}) + P(\text{chien}|\text{dog})$

$= 1$

$i = 1$ $tc(f_2|e_1) = 0 + \frac{P(f_2|e_1)}{\text{Total}} = 0 + \frac{\frac{1}{3}}{1} = 0.33$

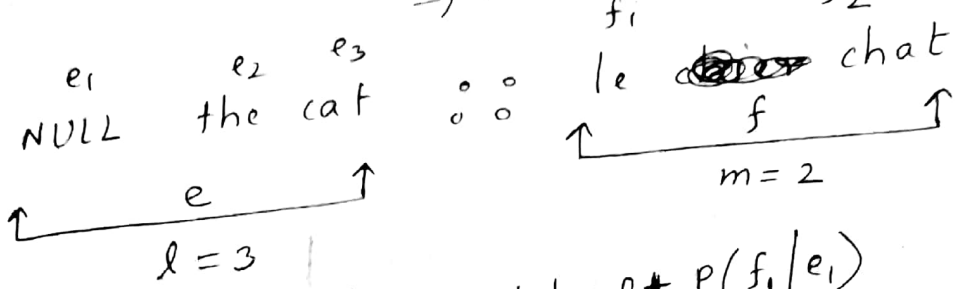
$\Rightarrow tc(\text{chien}|\text{NULL})$

$i = 2$ $tc(f_2|e_2) = 0 + \frac{P(f_2|e_2)}{\text{Total}} = 0 + \frac{\frac{1}{3}}{1} = 0.33$

$\Rightarrow tc(\text{chien}|\text{the})$

$i = 3$ $tc(f_2|e_3) = 0 + \frac{P(f_2|e_3)}{\text{Total}} = 0 + \frac{\frac{1}{3}}{1} = 0.33$

$\Rightarrow tc(\text{chien}|\text{dog})$



$j = 1$ total = 0 $i = 1$ total = $0 + P(f_1|e_1)$

$i = 2$ total = $P(f_1|e_1) + P(f_1|e_2)$

$i = 3$ total = $P(f_1|e_1) + P(f_1|e_2) + P(f_1|e_3)$

$= P(\text{le}|\text{NULL}) + P(\text{le}|\text{the}) + P(\text{le}|\text{cat})$

$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$

$i = 1$ $tc(\text{le}|\text{NULL}) = 0.333 + \frac{P(\text{le}|\text{NULL})}{\text{total}} = 0.333 + 0.666$

$i = 2$ $tc(\text{le}|\text{the}) = 0.333 + \frac{P(\text{le}|\text{the})}{\text{total}} = 0.333 + 0.666$

$i = 3$ $tc(\text{le}|\text{cat}) = 0 + \frac{P(\text{le}|\text{cat})}{\text{total}} = 0 + 0.$

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$j=2$ total $=0$ $i=1$

$i=2$ total $= P(f_2|e_1) + P(f_2|e_2)$

$i=3$ total $= P(f_2|e_1) + P(f_2|e_2) + P(f_2|e_3)$

$= P(\text{chat}|\text{NULL}) + P(\text{chat}|\text{the}) + P(\text{chat}|\text{cat})$

$= 1$

$i=1$ $tc(\text{chat}|\text{NULL}) = 0 + \frac{P(\text{chat}|\text{NULL})}{\text{total}} = 0 + \frac{1/3}{1} = 0.333$

$i=2$ $tc(\text{chat}|\text{the}) = 0 + \frac{P(\text{chat}|\text{the})}{\text{total}} = 0 + \frac{1/3}{1} = 0.333$

$i=3$ $tc(\text{chat}|\text{cat}) = 0 + \frac{P(\text{chat}|\text{cat})}{\text{total}} = 0 + \frac{1/3}{1} = 0.333$

(Recompute translation probabilities)

$\text{total}(\text{the}) = 0 + tc(\text{le}|\text{the}) + tc(\text{chien}|\text{the}) + tc(\text{chat}|\text{the})$

$= 0.666 + 0.333 + 0.333$

$= 1.333$

$P(\text{le}|\text{the}) = \frac{tc(\text{le}|\text{the})}{\text{total}(\text{the})} = \frac{0.666}{1.333} = 0.5$

$P(\text{chien}|\text{the}) = \frac{tc(\text{chien}|\text{the})}{\text{total}(\text{the})} = \frac{0.333}{1.333} = 0.25$

$P(\text{chat}|\text{the}) = \frac{tc(\text{chat}|\text{the})}{\text{total}(\text{the})} = \frac{0.333}{1.333} = 0.25$

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total (dog):

$$total(dog) = 0 + tc(le|dog) + tc(chien|dog)$$
$$= 0 + \frac{1}{3} + \frac{1}{3}$$
$$= 0.666$$

$$P(le|dog) = tc(le|dog) / total(dog)$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

$$P(chien|dog) = tc(chien|dog) / total(dog)$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

total (cat):

$$total(cat) = 0 + tc(le|cat) + tc(chat|cat)$$
$$= 0.333 + 0.333$$
$$= 0.666$$

$$P(le|cat) = \frac{tc(le|cat)}{total(cat)}$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

$$P(chat|cat) = \frac{tc(chat|cat)}{total(cat)}$$
$$= \frac{0.333}{0.666}$$
$$= 0.5$$

total (NULL):

$$total(NULL) = 0 + tc(le|NULL) + tc(chien|NULL) + tc(chat|NULL)$$
$$= 0.666 + 0.333 + 0.333 = 1.332$$

$$P(le|NULL) = \frac{tc(le|NULL)}{total(NULL)} = \frac{0.666}{1.332}$$

$$P(chien|NULL) = \frac{tc(chien|NULL)}{total(NULL)} = \frac{0.333}{1.332}$$

$$P(chat|NULL) = \frac{tc(chat|NULL)}{total(NULL)} = \frac{0.333}{1.332}$$

iteration 2:

Go to step 3

iteration 3:

Go to step 3

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